

ARUN KUMAR S

Full Stack Developer | Angular | .NET Web API | Public Safety Software
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PROFESSIONAL SUMMARY

Full Stack Developer with 3+ years of experience building scalable web, mobile, and enterprise applications in the Public Safety Software domain. Specialized in Angular, .NET Web API, and Python/ML with hands-on expertise in facial recognition, real-time communication via SignalR, secure payment integrations (Authorize.net, Muncipay), and AI-powered analytics. Proven track record of delivering mission-critical features, resolving production issues, and leading cross-functional teams in Agile environments.

PROFESSIONAL EXPERIENCE

Full Stack Developer – Public Safety Software | Prophoenix Technologies, Chennai Mar 2023 – Present

- Architected and deployed a cross-platform biometric security mobile application for public safety agencies using Angular 20 and Apache Capacitor, integrating Python-based facial recognition microservices to authenticate 10,000+ users with 99.2% accuracy.
- Engineered a high-performance real-time messaging application using Angular SignalR and .NET Web API, reducing message latency by 40% and sustaining 99.9% uptime across 50,000+ concurrent connections in public safety command environments.
- Implemented Authorize.net and Muncipay payment gateways with PCI-DSS Level 1 compliance, processing \$500K+ monthly transactions with 99.97% uptime and zero security incidents.
- Built Python-based analytics solution using NumPy and Pandas for proprietary public safety database analysis, improving data processing efficiency by 35% and enabling automated predictive insights for law enforcement decision-making.
- Led a cross-functional team of 5 developers using Agile/Scrum methodologies, resolved 25% more customer issues than the previous cycle, and mentored junior developers on coding best practices.
- Developed and maintained enterprise CRM solutions for 500+ public safety clients using Angular and .NET Web API, implementing UI/UX enhancements that increased user adoption by 18%.
- Reduced bug-related support tickets by 25% through systematic debugging, root cause analysis, and collaboration with the support team.
- Improved project delivery timelines by 15% through effective sprint planning, time management, and continuous integration practices using Git and Docker.

KEY PROJECTS

Enterprise Biometric Security Application

2024

Tech: Angular 20, Apache Capacitor, .NET Web API, Python ML

- Built a full-stack secure mobile application with facial recognition for enterprise access control in the public safety domain.
- Integrated Python OpenCV and TensorFlow models via RESTful microservices, achieving 99.2% recognition accuracy with real-time processing under 2 seconds.
- Implemented local SQLite caching for offline functionality, reducing API calls by 30% in low-connectivity field environments.
- Deployed cross-platform app for iOS and Android using Apache Capacitor with native camera integration.

Real-Time Collaboration & Messaging Platform

2023–2024

Tech: Angular, SignalR, .NET Web API

- Architected SignalR hub infrastructure supporting 50,000+ concurrent WebSocket connections with horizontal scaling for public safety operations centers.
- Reduced message latency from 800ms to 480ms (40% improvement) via WebSocket optimization and connection pooling.
- Implemented AES-256 end-to-end message encryption and comprehensive audit logging to meet enterprise compliance requirements.
- Built notification system with offline message queuing and delivery confirmation tracking.

Payment Gateway Integration Suite2023

Tech: Authorize.net, Municipay, .NET API, C#

- Integrated dual payment gateways with secure tokenization, encrypted data transmission, and PCI-DSS Level 1 compliance.
- Processed \$500K+ monthly transactions with 99.97% uptime using retry logic and fallback mechanisms.
- Developed custom fraud detection logic via transaction pattern analysis, reducing chargebacks by 12%.
- Built a payment reconciliation dashboard with automated reporting using Entity Framework and MS SQL Server.

AI-Powered Analytics Dashboard2024

Tech: Python, NumPy, Pandas, PostgreSQL, Angular, Chart.js

- Built an automated data processing pipeline to extract, transform, and analyze public safety data from proprietary databases.
- Implemented statistical analysis and predictive modeling, reducing manual analysis time by 35%.
- Created an interactive visualization dashboard for non-technical stakeholders using Angular and Chart.js.
- Trained a cross-functional team of 8 on data science best practices and self-serve analytics tools.

TECHNICAL SKILLS

Frontend	Angular (v2-20), TypeScript, RxJS, NgRx, HTML5, CSS3, SASS, JavaScript
Mobile	Ionic Framework, Apache Capacitor (iOS / Android cross-platform)
Backend	.NET Core, .NET 8, C#, ASP.NET Web API, SignalR, Entity Framework Core, RESTful APIs
AI & ML	Python, NumPy, Pandas, Scikit-learn, OpenCV, TensorFlow, Face Recognition
Databases	MS SQL Server, PostgreSQL, MongoDB, SQLite
Payments	Authorize.net, Municipay, PCI-DSS Compliance
DevOps	Git, GitHub, Docker, Jira, Visual Studio, VS Code
Methodology	Agile, Scrum, TDD, CI/CD

EDUCATION

Bachelor of Engineering in Computer ScienceGraduated 2022

Navalar Nedunchezhiyan College of Engineering, Cuddalore, Tamil Nadu

CERTIFICATIONS

Angular and Web API Full Stack Development Certification — Besant Technologies